

Lecture 5

14/08/24 9:53 AM

• RANDOM VARIABLES

We may be interested in functions of sample points than the samples themselves.

e.g. rolling a die 10 times

X : no. of times 6 appears

↳ random variable

- A random variable X is a function $X: \Omega \rightarrow \Omega'$ that transforms the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ to $(\Omega', \mathcal{F}', P_X)$ and is $(\mathcal{F}, \mathcal{F}')$ -measurable.

P_X = induced probability measure

$$X(\omega) \in \Omega' \quad \forall \omega \in \Omega$$

- For event $B \in \mathcal{F}'$ the pre-image $X^{-1}(B)$ is defined as $X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\}$

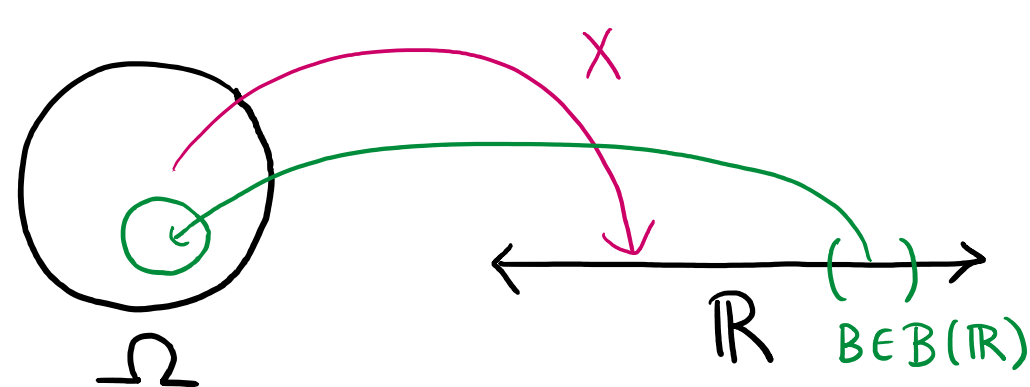
- $(\mathcal{F}, \mathcal{F}')$ -measurability
 $\Rightarrow$ for every $B \in \mathcal{F}'$, we have $X^{-1}(B) \in \mathcal{F}$.

Since $X^{-1}(B) \in \mathcal{F}$, it can be measured using $\mathbb{P}$.

$$P_X(B) := \mathbb{P}(X^{-1}(B)) \quad \forall B \in \mathcal{F}'$$

- If there is no $\omega \in \Omega$ such that $X(\omega) \in B$
 $\Rightarrow X^{-1}(B) = \emptyset$
 $\Rightarrow P_X(B) = \mathbb{P}(X^{-1}(B)) = \mathbb{P}(\emptyset) = 0$

$\therefore$ We can use $\Omega' = \mathbb{R}$
 $\mathcal{F} = \mathcal{B}(\mathbb{R})$ *borel sigma-algebra*



$$\Omega \xrightarrow{X} \mathbb{R}, \quad \mathcal{F} \xrightarrow{X} \mathcal{B}(\mathbb{R}), \quad \mathbb{P}(\cdot) \xrightarrow{X} P_X(\cdot)$$

- A random variable X is a map $X: (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}), P_X)$ such that for each $B \in \mathcal{B}(\mathbb{R})$, the inverse image $X^{-1}(B) := \{\omega \in \Omega : X(\omega) \in B\}$ satisfies:

$$X^{-1}(B) \in \mathcal{F} \text{ and } P_X(B) = \mathbb{P}(\omega \in \Omega : X(\omega) \in B)$$

RANDOM VARIABLES $\left\{ \begin{array}{l} \text{discrete} \Rightarrow \Omega' \text{ is countable} \\ \text{continuous} \Rightarrow \Omega' \subseteq \mathbb{R} \end{array} \right.$

• DISCRETE RANDOM VARIABLES

$$P_X(\{x\}) = \mathbb{P}(\{\omega \in \Omega : X(\omega) = x\}) \quad !!!$$

REMEMBER CURLY BRACES FOR SET FNS

• PROBABILITY MASS FUNCTION

$$= p_X(x) := P_X(\{x\}) \text{ for } x \in \Omega'$$

• CUMULATIVE DISTRIBUTION FUNCTION

$$= F_X(x) := \sum_{x \leq x_i} p_X(x) \quad \text{increasing curve}$$

• EXPECTATION / MEAN of a random variable

$$E[X] = \sum_{x \in \Omega'} x \cdot p_X(x)$$

• MOMENT of a random variable

$$n^{\text{th}} \text{ moment of } X = E[X^n] = \sum_{x \in \Omega'} x^n \cdot p_X(x)$$

- For a function $g(\cdot)$ of a RV X ,

$$E[g(x)] = \sum_{x \in \Omega'} g(x) \cdot p_X(x)$$

Functions of random variables are also random variables.

Normal mean

$$= \frac{x_1 + x_2 + \dots + x_{10}}{10} = x_1 \times \frac{1}{10} + x_2 \times \frac{1}{10} + \dots + x_{10} \times \frac{1}{10}$$

Assumes each value has equal probability (1/10).